Higher National Diploma in Engineering (Electrical)

First Year, 1St Semester Examination — 2020

EE 1107 Electrical Principals A

Question 1. (Total 20 Marks)

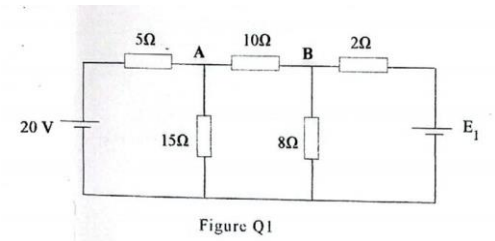

Figure Q1

The voltage between AB is 7V when AB is an open circuit. The resistance between A and B is 5.35Ω when AB is an open circuit and two emf sources are replaced by their internal resistances (here internal resistances are zero) in the circuit given in figure Q1.

i) Find the value of E1 (Hint: This is an application of Thevenin's theorem) [10 marks]

ii) Find the current through the 10Ω resistor at the circuit given in figure Q1 by applying the superposition theorem. [10 marks]

Question 2

i) Name three (03) types of fixed capacitors. [03 marks]

ii) Draw the charging current graph of a capacitor and roughly mark the time constant on the graph. [03 marks]

iii) A $1\mu\mathrm{F}$ capacitor is connected to a $50\ \mathrm{V}$ supply by closing a switch through a $2\ \mathrm{M}\Omega$ series resistor. Calculate the followings; [06 marks] (a). The time constant (b). Initial charging current (c). Voltage across the capacitor $5\ \mathrm{seconds}$ after closing the switch (d). The time taken by the capacitor for charging it fully

iv) A parallel plate Air capacitor is charged up to $Q\ \mathrm{coulombs}$. The Air gap of the same capacitor is filled with a dielectric of dielectric strength of 1.8. The energy stored in the latter capacitor is $3\ \mathrm{times}$ that of the aforesaid Air capacitor. Find the charge on the dielectric-filled capacitor. [06 marks]

v) A charge of $5\mu\mathrm{C}$ is placed in an electric field experiencing a force of $2.5\ \mathrm{N}$. What is the magnitude of electric field intensity? [02 marks]

Question 3

i) Draw the direction of self-induced emf at the inductor given in figure Q2, when the value of the variable resistor R is increased after closing the switch. [02 marks]

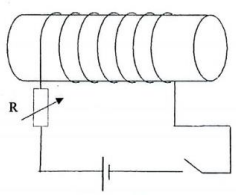

Figure - 02

ii) State the direction that a conductor is needed to be moved in a magnetic field, in order to produce the maximum emf. [02 marks]

iii) State the two ways of spending the energy supply to the inductive circuit. [04 marks]

iv) Two coils having 50 and $100\ \mathrm{turns}$ respectively are wounded side by side on an iron core and formed a closed circuit. Cross-sectional area and mean length of the core are $100\ \mathrm{cm}^2$ and $200\ \mathrm{cm}$ respectively. (a). Estimate the mutual inductance between the coils if the relative permeability of the iron is 2000. [03 marks] (b). Find the emf induce in the second coil if a current of zero Ampere grows to $20\ \mathrm{A}$ in a time of $0.02\ \mathrm{seconds}$ in the 1st coil. [03 marks]

v) A steel ring with a $20\ \mathrm{cm}$ mean radius and circular section of $2\ \mathrm{cm}$ diameter has an Air gap of $1\ \mathrm{mm}$ wide. It is wounded uniformly with $500\ \mathrm{turns}$ of wire carrying the current of $2\ \mathrm{A}$. If the air gap takes 70% of the total mmf, what is the reluctance of the complete magnetic circuit? (You may neglect the magnetic leakage.) [06 marks]

Question 4

i) Copy the sets of waveforms given in figure Q4 (i)(a) and figure Q4 (i) (b) and mark peak amplitudes, period, and phase difference. Sketch the phasor diagram of each set of waveforms. [04 marks]

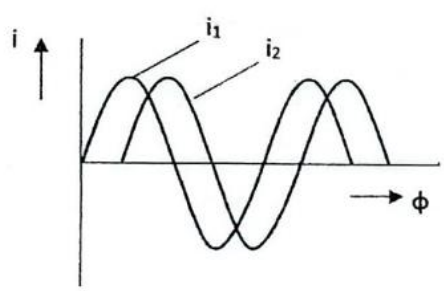

Figure – 04(i)(a)

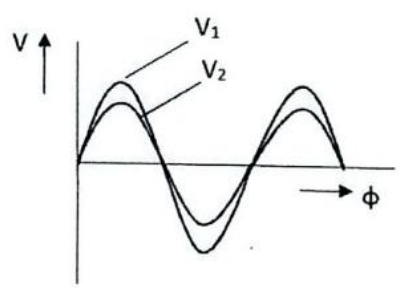

Figure – 04((i)(b)

ii) The frequency of the signal given below is $60\ \mathrm{Hz}$ and the peak amplitude is $5\ \mathrm{V}$. Find the rms voltage of the alternating voltage. Also, find the time period for completing one cycle. [03 marks]

iii) It is found that the voltage is leading the current by an angle Φ^o when an alternating voltage is applied to the particular circuit. Draw the voltage triangle and impedance triangle, and derive an expression for the power factor of the circuit. (Hint: the relationship between R and Z) [03 marks]

iv) A capacitor C is connected in series with a 40Ω resistor across a supply of frequency $60\ \mathrm{Hz}$. A current of $3\ \mathrm{A}$ flows and the circuit impedance is 50Ω. Calculate the following. [06 marks] (a). The value of the capacitor (b). The supply voltage (c). The phase angle between the supply voltage and current (d). The potential difference across the resistor (e). The potential difference across the capacitor (f). Draw the phasor diagram

v) A coil having a resistance of 10Ω and an inductance of $125\ \mathrm{mH}$ is connected in series with a $60\mu\mathrm{F}$ capacitor across a $120\ \mathrm{V}$ supply. At what frequency does the resonance occur? Find the current through the circuit at this frequency. [04 marks]

Question 5

Two impedance given by $Z_1 = (4 + j3)\Omega$ and $Z_2 = (8 - j6)\Omega$ are joined in parallel and connected across a voltage of magnitude 200 and angle 0° (i.e. $200\angle 0^\circ$). Calculate,

i) Circuit current [06 marks]

ii) Current in each branch [06 marks]

iii) Average power delivered to the circuit [02 marks]

iv) Power taken by each branch [02 marks]

v) Power factor of individual circuits and combination [04 marks]

Question 6

i) Write down two advantages of a three-phase power system over a single-phase power system. [02 marks]

ii) What is meant by 'balanced load' in a three-phase power system? [02 marks]

iii) State the possible phase sequences in a three-phase electrical power system if the three phases are $\mathrm{R}, \mathrm{Y}, \mathrm{B}$? [03 marks]

iv) Mark the line voltage and phase voltage in a three-phase star connection with the help of a diagram. [03 marks]

v) An impedance of $(12 + j16)\Omega$ balance three-phase delta connected load consumes $4.8\ \mathrm{kW}$ per phase. The line current drawn by the load is $34.64\ \mathrm{A}$. Find the line and phase voltages. [05 marks]

vi) A star-connected load consists of three (03) identical coils each of resistance 30Ω and inductance $127.3\ \mathrm{mH}$. When the line current is $5.08\ \mathrm{A}$ and the supply frequency is $50\ \mathrm{Hz}$, calculate the line voltage. [05 marks]